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House Veridian
The Drakkar Ascent Challenge

The Lightest Thrust Barrel Wins

Aircraft Structures Course Assignment-Competition

Third-Year Aerospace Engineering MDM

Semester V (Academic Year 2026-27)

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Course Code: R5ME3203T

Maximum Marks: 100

1 The Summons

In the era of the Eternal Imperium, the desert world of **Drakkar** has long been ignored by the great Houses. It is a scorched, wind-scoured planet, its only value the rich veins of **zhalite ore** beneath the sand — the rare mineral that powers interstellar drives.

House Veridian, a minor noble house fallen on hard times, has discovered a vast zhalite deposit in the equatorial wastes. To export the ore and restore the House's fortunes, they must build an **orbital lifter**: a reusable surface-to-orbit booster capable of surviving Drakkar's brutal atmosphere and carrying zhalite to the Guild freighters in orbit.

But the Imperium's tariffs are merciless. Every kilogram of structural mass on the lifter reduces the zhalite payload. Every unnecessary margin is lost wealth. Every design flaw means ruin.

Between the propellant tank and the engine thrust structure sits the most unforgiving component: the **thrust barrel**.

It carries the full thrust of the ascent engine. It bends under Drakkar's violent thermal gusts. It resists the tank's internal pressure. And if its wall is too thin, it buckles catastrophically, killing the crew and burying the House's hopes in the sand.

The current metallic design is robust. It is also far too heavy.

The Lord of House Veridian has called a design competition among the nine (or so) engineering cadres of the House academy. One winner. One design. The lightest thrust barrel that survives every load case will be built and flown on the first cargo ascent.

Welcome to the Drakkar Ascent Challenge.

2 The Charge

Design, analyse, and optimise a **cylindrical booster thrust barrel section** for the Veridian orbital lifter.

The structure must:

- fit within a **1.5 m length** and **300 mm maximum outer diameter**,
- withstand the loads that arise from the flight scenario described below,
- avoid material failure,
- avoid global column buckling,
- avoid local shell buckling,
- minimise total structural mass.

This is a **purely theoretical** design competition. No physical prototype. No finite element analysis.

3 The Trial by Sky

The Veridian lifter is ascending through Drakkar's lower atmosphere at the instant of maximum dynamic pressure. The thrust barrel is the cylindrical section between the propellant tank and the engine thrust structure.

At this instant, the following physical conditions apply.

3.1 Engine and Tank Conditions

- The ascent engine is producing a vacuum-equivalent thrust of $T = 30\,\text{kN}$. This thrust is transmitted axially through the thrust barrel.
- The propellant tank ullage pressure is $p_u = 0.3\,\text{MPa}$. This pressure acts on the inside wall of the thrust barrel.

3.2 Thermal Gust Conditions

Drakkar's sun-baked sands generate violent rising thermals. As the lifter climbs through the desert atmosphere, it encounters a sharp vertical gust. The resulting side force on the upper vehicle must be carried by the thrust barrel at its base.

The relevant parameters are:

Parameter	Symbol	Value
Flight speed	V_∞	$200\,\text{m/s}$
Atmospheric density	ρ	$1.225\,\text{kg/m}^3$
Vertical gust velocity	w_g	$20\,\text{m/s}$
Upper vehicle length above thrust barrel	L_u	$3.0\,\text{m}$
Upper vehicle diameter	D_u	$0.10\,\text{m}$
Normal-force coefficient slope	$C_{N\alpha}$	$2.0\,\text{rad}^{-1}$

Assume the upper vehicle can be modelled as a uniform slender cylinder of diameter D_u and length L_u. The aerodynamic side force per unit length is constant along this length.

4 Deriving the Loads

You must compute the loads acting on the thrust barrel from the scenario above.

4.1 Dynamic Pressure

$$q = \frac{1}{2}\,\rho\,V_\infty^2 \tag{1}$$

4.2 Equivalent Angle of Attack due to Gust

$$\alpha = \arctan\left(\frac{w_g}{V_\infty}\right) \tag{2}$$

4.3 Distributed Side Force per Unit Length

For a slender cylinder at angle of attack α:

$$f = q\,C_{N\alpha}\,\alpha\,D_u \tag{3}$$

4.4 Total Side Force on Upper Vehicle

$$F_{side} = f\,L_u \tag{4}$$

4.5 Shear Force at the Thrust Barrel

$$V = F_{side} \tag{5}$$

4.6 Bending Moment at the Thrust Barrel

For a uniform distributed load over length L_u:

$$M = \frac{f\, L_u^2}{2} \tag{6}$$

4.7 Axial Compressive Force

$$F = T = 30\,\text{kN} \tag{7}$$

4.8 Internal Pressure

$$p = p_u = 0.3\,\text{MPa} \tag{8}$$

Use the values you compute for F, M, p, and V in all subsequent strength and buckling checks.

5 The Materials

You may choose from advanced metals, quasi-isotropic composites, or a hybrid combination.

5.1 Metallic Materials (Isotropic)

Material	E **(GPa)**	σ_y **(MPa)**	ρ **(kg/m^3)**	ν
Aluminium 7075-T6	71.7	503	2810	0.33
Titanium Ti-6Al-4V	113.8	880	4430	0.31
Steel 4130	205	460	7850	0.29
Magnesium AZ31B	45	220	1770	0.35

5.2 Quasi-Isotropic Composite Laminates

Composite System	**Effective** E **(GPa)**	ν	X_c **(MPa)**	Y_t **(MPa)**	S **(MPa)**	ρ **(kg/m^3)**
Carbon/Epoxy QI	55	0.30	600	500	200	1600
Glass/Epoxy QI	25	0.30	300	250	100	1900
Aramid/Epoxy QI	35	0.34	200	300	80	1400

Composite allowables are first-ply failure values under uniaxial loading. Hybrid designs are permitted, but the analysis must be transparent.

6 The Analysis

No FEA. No black-box software.

You will use:

- thin-walled cylinder theory,
- von Mises yield criterion for metals,
- a quadratic interaction criterion for composites,
- Euler column buckling,
- local shell buckling with knockdown factors,
- mass optimisation through parametric trade studies.

7 The Rules of Engagement

Parameter	Requirement
Length	1500 mm
Maximum outer diameter	300 mm
End conditions	Simply supported rigid end rings
Load introduction	Uniform axial compression and bending at ends
Minimum wall thickness	No formal minimum, but must be manufacturable and realistic
Stiffeners	Permitted, but local panel buckling must be checked
FEA	Not permitted
Experimental testing	Not required
Team size	Maximum 5

8 The Reckoning

The winning cadre will not be the one with the thickest wall. The winning cadre will be the one that understands exactly how thin the wall can be before failure.

Performance Score

Let m_i be the predicted structural mass of team i.

A design either passes or fails.

- **Pass:** all strength and buckling checks satisfied with required factors of safety.
- **Fail:** any check violated.

For passing designs only:

$$S_i = \frac{m_{\min}}{m_i}$$

where $m_{\min}$ is the lightest passing design in the competition.

Total Score

$$\text{Total}_i = 100 \times [0.50\, S_i + 0.25\, R_i + 0.15\, T_i + 0.10\, I_i]$$

where:

- R_i = design report quality,
- T_i = theoretical correctness and clarity,
- I_i = innovation and manufacturability.

9 The Timeline

Week	Deliverable
Week 1	Preliminary design: material selection, initial sizing, hand calculations, concept sketches
Week 2	Detailed analysis: stress, failure, buckling checks, parametric trade study
Week 3	Final optimised design, mass budget, all calculations, drawings
Week 4	House Veridian design review board presentation and peer review

10 Final Briefing

Your cadre has four weeks.

The flight scenario is fixed. The materials are defined. The physics is unforgiving.

Derive the loads. Design the lightest thrust barrel that survives.

May the desert winds favour the light.